Dual-System VLA on LIBERO

Code, videos and experiment log: github.com/tkpf/dual-system-architecture-vla

Task

A Dual-System policy for the LIBERO benchmark is to be built. System 2 (slow, semantic) conditions System 1 (fast, reactive). An ablation must show whether System 2 really improves System 1. Hi Robot [1] served as role model: a high level writes text commands, a low-level VLA executes them.

Key results at a glance	
Counterfactual	Removing System 2 does not hurt: 9/20 without it, 4/20 with it (`libero_10`). Freezing it hurts (1/4).
Error recovery	Drops are detected in 8/8 cases. Success is unchanged with and without recovery (6/8 each).
Bottleneck	Execution, not reasoning: 13 of 16 failures also fail under oracle switching.

1 Prerequisites

- *No private GPU.* Surface Laptop Studio, Intel Iris Xe, 16 GB RAM, no CUDA.
- *One week* for design, experiments and report.
- *Little experience* with VLMs and VLAs.

2 Tools

- *Google Colab:* T4 and L4 GPUs for all evaluation.
- *Colab MCP server:* lets the coding agent drive a Colab notebook.
- *Coding LLMs:* Claude Sonnet 5.5 and Opus 5.5.

Models used and investigated

Model	Role	Checkpoint	Status
pi0.5	System 1, final	`lerobot/pi05_libero_finetuned_v044`	Used frozen. Finetuned for 400 steps: no gain
SmolVLA	System 1, v1	`lerobot/smolvla_libero_plus` (finetuning started from `lerobot/smolvla_libero`)	Replaced by pi0.5. Finetuned for 600 steps: no gain
Qwen3-VL 2B	System 2 VLM, final	`Qwen/Qwen3-VL-2B-Instruct`	Used frozen, for act phases only
Qwen3-VL 4B, 8B	Offline judge tests	`Qwen/Qwen3-VL-4B-Instruct`, `Qwen/Qwen3-VL-8B-Instruct`	Investigated: not reliable enough
SmolVLM2 2.2B	Offline judge tests	`HuggingFaceTB/SmolVLM2-2.2B-Instruct`	Investigated: not better

Table 1: Models. Software: LeRobot 0.6.1 with the official LIBERO pipeline.

3 Architecture

3.1 Generic design: frozen VLM and VLA, joined by text

- *Both models are frozen.* System 2 writes *one English sentence* into System 1's prompt.
- Interpretable. The interface is readable text. No latent embedding sits in between.
- *Easy to ablate.* System 2 can be removed, frozen, replaced by an oracle or corrupted. Nothing is retrained.
- *No training needed.* This fits the compute and time budget.
- *Price:* neither side can adapt to the other. Section 3.2 shows where this limit bit.

3.2 How the implementation evolved

The generic design never changed. Two things changed: *when* System 2 runs and *who judges progress.*

Version	System 2 runs	Progress judged by	Outcome
v1: SmolVLA, VLM judge	Every 20 steps, plus environment triggers (gripper toggle, robot stops, empty grasp, 300-step timeout)	Qwen3-VL 2B: running, done or failed. "Success" ends the episode	Success 19/36 → 4/36. 7 of 11 success verdicts were wrong
v2: pi0.5 as System 1	Same triggers	Same VLM judge	pi0.5 follows commands (16/17 against 3/17 for gibberish). Dual still worse: 8/12 → 3/12
v3: phase-aware prompt	Triggers, 150-step timeout	VLM told phase and "done" criteria, short reasoning	No false successes, but 0 "done" in 76 checks
v4: final	*Every step* on robot signals. VLM every 10 steps, act phases only	*Gripper width and hand height*; VLM only for "turn on", "open", "close"	Matches oracle switching. Never ends an episode

Table 2: Evolution of System 2. The generic design is identical in all rows.

What the failures showed:

1. *SmolVLA ignores language.* Gibberish gave the same behaviour as real commands. pi0.5 replaced it.

2. *Finetuning System 1 on subtasks did not help.* pi0.5 reached 17/28 goals against 21/28 stock. Both policies stay frozen.
3. The VLM judge is the weak part. No untrained VLM up to Qwen3-VL 8B judged "finished" reliably (best AUROC 0.71, about 1/3 false "done"). Qwen 2B calls a glowing burner dark.
4. *Hence v4:* the progress check moved to robot signals. The VLM is kept only where no signal exists.

3.3 Final architecture

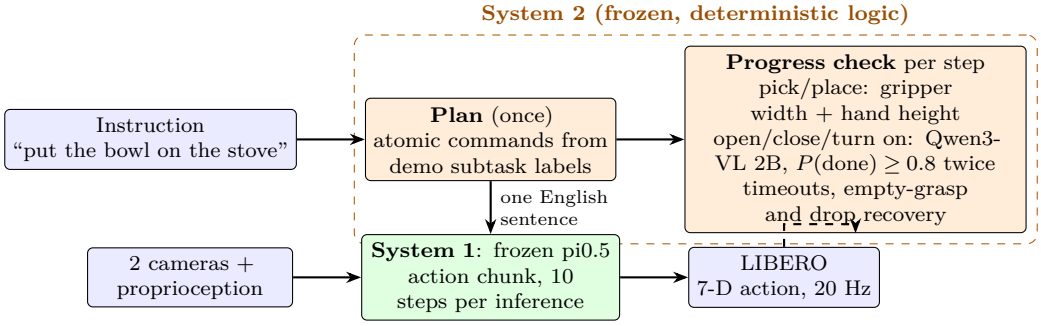


Figure 1: Final system. A changed command clears System 1's queued actions. Dashed: gripper and camera signals feed the progress check.

- *Pick done:* gripper holds something for 8 steps and the hand rose 3 cm.
- *Place done:* gripper reopens after a hold.
- *Timeout:* 150 steps per phase.
- *Empty grasp:* "open the gripper", then retry.

4 Ablation

Protocol: same scene, robot start and seed for every condition. Shifted starts move the gripper by about 10 cm (C) or 19 cm (B).

Condition	Meaning
passthrough	System 2 removed. The whole instruction is the prompt.
frozen	System 2 frozen. First atomic command for the whole episode.
fused	Final System 2.
oracle	Simulator decides when a phase is done. Upper bound for any progress check.

4.1 Counterfactual: is System 2 needed?

Condition	`libero_goal`, 4 runs (median steps)	`libero_10`, 10 scenes × 2 seeds, start B	Drop test, 8 episodes
passthrough	4/4 (99)	9/20	5/8
frozen	1/4 (218)	7/20	—
fused	4/4 (101)	4/20	6/8
oracle	4/4 (102)	5/20	5/8

Table 3: Success counts. Drop test: section 4.2.

- Frozen System 2 fails: the bowl is held, never placed (1/4).
- Fused ≈ oracle: switches 0–2 steps from simulator truth. Same outcome in 15 of 20 pairs.
- System 2 does not beat passthrough. Oracle also drops to 5/20.
- *Likely cause:* bare atomic commands are out of distribution for pi0.5. It was trained on whole instructions.
- *Significance:* none. Passthrough against fused: 7 against 2 pairs, p=0.18.
- *Consistency:* both seeds agree in 9/10 scenes with the whole instruction, in 5–6/10 with atomic commands.

4.2 Error recovery

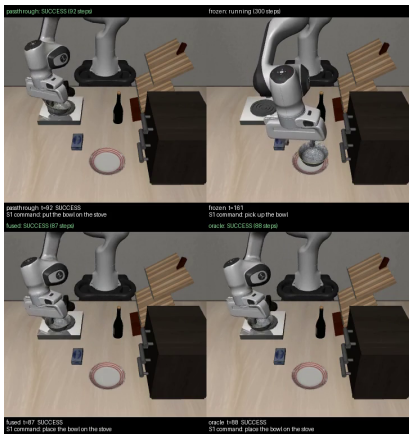
Perturbation: after 20 steps of holding, the gripper is forced open for 10 steps. The bowl falls. Two bowl scenes, start C, 4 seeds.

- Detection works: 8 of 8 drops found within 1–3 steps of the simulator.
- Recovery changes nothing: fused with and without recovery fail on the same episodes (6/8 each).
- *Oracle recovery is no better:* 5/8.
- *Reason:* pi0.5 picks a dropped bowl up again by itself, even under "place".

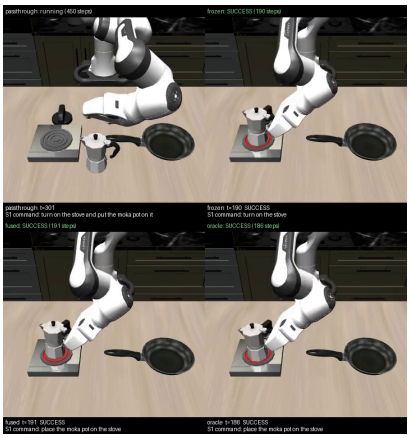
4.3 Bottleneck: reasoning or execution?

Method: each failed fused episode is checked against oracle switching on the same scene, start and seed.

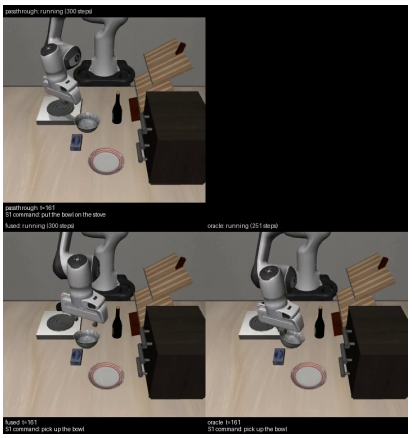
Failure source	Count	Evidence
Execution (System 1)	13 of 16	Oracle fails too. No better progress check helps.
System 2 or rollout noise	3 of 16	Oracle succeeds. One case inspected (book scene): switches 1 step apart, outcome flipped in the other seed.



Video 1: easy scene



Video 2: stove, then moka pot



Video 3: forced drop

Figure 2: Frames from the 2×2 rollout videos (top: passthrough, frozen; bottom: fused, oracle; each panel shows the command System 2 sends). (1) Frozen holds the bowl and never places it. (2) Atomic commands succeed where the whole instruction fails. (3) Fused notices the drop and re-commands the pick (frozen panel empty, not run). Captions link to the videos.

5 Limitations and conclusion

5.1 Limitations

- *Small samples:* 20 scene-seed pairs, 8 drop episodes, one init state.
- *Out-of-distribution sensitivity:* results depend on the robot start and the seed. Another robot or scene would likely behave differently.
- *System 1 follows language only weakly.* The open-source VLAs probably overfit LIBERO and generalize poorly. This is a likely cause, not a tested one.
- *System 2 was not fully exercised:* the plan came from demo subtask labels. Qwen's own decomposition matched the reference in only 3 of 4 `libero_goal` and 4 of 10 `libero_10` episodes. Reliable verbal task splitting by a VLM is therefore not shown.

5.2 Conclusion

- The frozen, text-interfaced Dual System is fully interpretable.
- System 2 matches an oracle at command switching and at drop detection.
- An improvement over System 1 alone could not be shown.
- With pi0.5, execution under atomic commands is the bottleneck, not reasoning.
- System 2 could not be used as intended. It was engineered for switching accuracy, and the true subtasks were injected in the final run.

5.3 What I would do next

- *Without training budget:* the choice was frozen models, because training outcomes are risky.
- *With more time and compute:* a diffusion policy that draws (dashed) bounding boxes around objects and place locations. These would be fed into a finetuned VLA.
- *Expected gain:* a more precise System 1 response.
- *Risk:* the VLA may rely on the boxes and obey language even less. The boxes can also be seen as a redundant second channel.

References

[1] L. X. Shi et al. *Hi Robot: Open-Ended Instruction Following with Hierarchical Vision-Language-Action Models*, arXiv:2502.19417, 2025.